CO2 - AT3

Class Test

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Course Code :- ITA0501

Course Name :- Computer vision.

slot :- A

1,A) Grey-level Transformations (log and power-Law)

Grey-level transformation is an image enhancement technique used to improve the brightness and visibility of dark regions.

1. Log Transformation:-

$$S = c \log(1+r)$$

where:

r = input pixel intensity.

S = output pixel intensity.

c = scaling constant.

2, Power-law (Gamma) Transformation:-

$$S = cr^{\gamma}$$

→ If $\gamma < 1$ the image becomes brighter.

→ If $\gamma > 1$ the image becomes darker.

→ If $\gamma = 1$ brightness remains unchanged.

Processing steps:-

1. Read the input image.
2. Convert the image into grayscale.
3. Apply log (or) gamma transformation.
4. Normalize the pixel values.
5. Display the enhanced image.

2)A) 2 Histogram Equalization:-

Histogram equalization is an image enhancement technique used to improve image contrast by spreading pixel intensities over a wide range.

It is useful for fingerprint images where pixel values are concentrated in a narrow intensity range.

$$S_k = (L-1) \sum_{j=0}^{k} P(r_j)$$

where:

→ L = Number of intensity levels.

→ $P(r_j)$ = probability of intensity level.

→ S_k = new intensity value.

Processing steps:-

1) Read the fingerprint image.
2) Convert it into grayscale.
3) Calculate the histogram.
4) Calculate the probability of each intensity.
5) Find the cumulative distribution function (CDF).
6) map old intensity values to new values.
7) Display the equalized image.

Result:- The contrast between fingerprint ridges and valleys increases, making fingerprint patterns clearer.

3,A) Salt-and-pepper Noise removal using median filter:-

Salt-and-pepper noise is a type of image noise that appears as random white and black pixels.
A median filter is suitable for removing this noise while preserving object edges.

Formula:-

$$g(x,y) = \text{median}\{f(s,t)\}$$

where:-

- $f(s,t)$ = pixels in the neighbourhood.
- $g(x,y)$ = output pixel value.

Processing steps:-

1. Read the noisy image.
2. select a 3x3 neighbourhood.
3. Arrange the pixel values in ascending order.
4. Find the middle(median) value.
5. Replace the centre pixel with the median.
6. Repeat for all pixels.

Why median filter instead of Averaging filter?
→ median filter removes extreme noise values.
→ It preserves edges better.
Result:- Noise is reduced while important edges and image details are preserved.

flat regions become smaller.

4A) Image sharpening using Laplacian Filter (or) Unsharp masking:

Image Sharpening is used to improve the edges and fine details of an image. In a printed circuit board image, sharpening helps to identify copper tracks clearly.

1. Laplacian filter:-

The laplacian is a second-order derivative operator used to detect rapid intensity changes.

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

Laplacian mask:-

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

2) unsharp masking:-

1) Blur the original image.
2) Subtract the blurred image from the original.
3) Obtain the mask.
4) Add the mask to the original image.

$$g = f + k(f - \bar{f})$$

Result:- copper tracks and edges become sharper and easier to inspect.

5.A) Frequency-Domain Low-Pass filtering :-

A low-pass filter is used to remove high-frequency noise and smooth an image. It is useful for reducing grainy noise in drone images of farm plots.

Types of Low-pass filters:-

1. Ideal Low-pass filter (ILPF)
2. Butterworth Low-pass filter (BLPF)
3. Gaussian Low-Pass Filter (GLPF)

Gaussian Low-pass filter formula:-

$$H(u,v) = e^{-D^2(u,v)/2D_0^2}$$

where:

→ $D(u,v)$ = distance from frequency origin.

→ D_0 = Cutoff frequency.

Processing steps:-

1) Read the drone image.
2) Convert the image into grayscale.
3) Apply fourier Transform.
4) Design a low-pass filter.
5) Multiply the frequency image with the filter.
6) Apply inverse fourier Transform.
7) Obtain the smoth image

Result:- High frequency noise is reduced, and farm plot regions become smoother.

6,A) Frequency-Domain High-Pass and High-boost filtering :-

A high-pass filter allows high-frequency components and removes low-frequency components. It is used to enhance edges and fine details in retinal fundus images.

High-pass filter formula:-

$$H_{HP}(u,v) = 1 - H_{LP}(u,v)$$

where:

→ H_{HP} = high-pass filter.

→ H_{LP} = Low-pass filter.

High-boost filtering :-

High-boost filtering enhances the details while retaining the original image information.

$$g(x,y) = Af(x,y) - f_L(x,y)$$

where :-

→ A = amplification factor.

→ $f(x,y)$ = original image.

→ $f_L(x,y)$ = Low-pass filtered image.

Preprocessing steps:-

1) Read the retinal fundus image & convert into grayscale
2) Apply fourier Transform.
3) Design a high-pass filter.

5) multiply the filter with the frequency spectrum.
6) Apply Inverse fourier transform.
7) Display the enhanced image.

Result:- Thin blood vessels, edges and fine details become more visible.

7A) Point detection and Line detection

Point detection and line detection are image processing techniques used to identify small isolated points and thin lines.

In glass-bottle inspection:

→ Point detection identifies tiny air bubbles.
→ Line detection identifies hairline cracks.

Processing steps:-

1) Read the bottle image.
2) Convert it into gray scale.
3) Apply the point detection mask.
4) Apply horizontal and vertical line masks.
5) calculate the response values.
6) Apply a suitable threshold.
7) Identify bubbles and cracks.

Result:- Tiny air bubbles and hairline cracks are identified on the bottle surface.

8,A) Image Thresholding (Global and Adaptive)

Thresholding is a image segmentation technique used to seperate foreground objects from the background.

1) Global Thresholding:-

$$g(x,y) = \begin{cases} 255 & f(x,y) \geq T \\ 0 & f(x,y) \leq T \end{cases}$$

2) Adaptive Thresholding:-

Processing steps:-

1) Read the handwritten image.
2) Convert it into grayscale.
3) Apply global (or) adaptive thresholding.
4) Seperate dark handwriting from the background.
5) Display the binary image.

Why adaptive thresholding:-

→ Global thresholding uses one threshold for the entire Image.
→ Adaptive thresholding uses local threshold values.
→ It works better when lighting is uneven across the paper.

Result:-
Handwritten text is seperated clearly from the white background.

9,A) Region growing and Region splitting and merging:-

Region-based segmentation is used to divide an image into meaningful regions based on pixel similarity.

1. Region growing:-

Processing steps:-

1) Read the satellite image.
2) Convert it into grayscale.
3) Select a seed point inside the lake.
4) Check neighbouring ~~skill~~ pixels.
5) Compare their intensity values with the seed region.
6) Add similar pixels to the region.
7) Repeat until no more similar pixels are found.

2) Region splitting and merging:-

→ splitting divides an image into smaller regions when pixels are not similar.

Similarity criterion:-

$$|I(x,y) - I_s| < T$$

Result:- The lake region is extracted separately from the surrounding land.

10)A) Edge Linking and Boundary detection:-

Edge linking is a technique used to Connect broken (or) discontinuous edges in an image.

1. Local processing:-

Local processing connects near by edge pixels based on:

1) pixel Intensity.
2) Gradient magnitude.
3) Gradient direction.

2) Hough transform:-

$$\rho = x\cos\theta + y\sin\theta$$

3) Boundary detection:-

Boundary detection identifies the outer edges of an object.

Processing steps:-

1) Read the mechanical gear image.
2) Convert into gray scale & Apply edge detection.
3) Identify broken edge points.
4) Apply local edge linking (or) Hough Transform.
5) Connect discontinuous edges.
6) Extract the complete gear Boundary.

Result: Broken edges are connected & a complete mechanical gear is obtained.